ENVIRONMENTAL PRODUCT DECLARATION



Archi+ Pietra Intonaco



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EPD of multiple products (Archi+ Pietra Intonaco, Archi+ Pietra, Archi+ Argilla, Archi+ Tadelakt, Archi+ Fondo, Archi+ Big and Archi+ Concrete), based on a representative product (Archi+ Pietra Intonaco)

An EPD should provide current information and may be updated if conditions change. The stated validity is therefore subject to the continued registration and publication at www.environdec.com.

DESIGN THE FUTURE

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1. The company and the product

San Marco Group SpA is working on the quantification of the environmental impacts of its products in order to improve their technical and environmental performance.

1.1 San Marco Group S.p.A.

San Marco Group is an international group, operating in the production and marketing of paints and varnishes for professional construction in Italy. The headquarters are located in Marcon, Venice, where part of the production takes place, together with the innovation centre and two training centres. The Group consists of 7 production sites and 3 sales companies located in Europe, where more than 300 people work directly. International growth has allowed the Group to be present in most parts of the globe with a network of distributors touching more than 100 countries on five continents, with a widespread network of sales outlets.



The Group manages 6 different brands. The key factors for San Marco Group are the search for opportunities in foreign markets, the acquisition of other entities and the strengthening of the services offered.

The Group's various brands, acquired or developed internally, have expanded business opportunities into new markets and new customer segments. All of the Group's companies have a precise role to play, so as to enhance their specificities and define their positions, reducing overlaps and exploiting synergies. This type of organisation has progressively led to the upgrading of machinery and plants, increasing the quantity and quality of products, all through a commitment to constant improvement and growth.

San Marco Group has voluntarily chosen to establish, implement, maintain and improve its integrated management system for quality, the environment and the health and safety of workers.

Indeed, San Marco Group obtained the ISO 9001:2015 Quality Certification through the Certification Body DNV-GL, for all the Italian production unit of San Marco Group S.p.A. (Marcon, Latisana and Montemarciano), for the company of Forlì, Novacolor S.r.l., for the Bosnian company San Marco DT BH Doo, and for the Norwegian company Tjaeralin AS. It obtained the ISO 14001:2015 Environmental Certification again with the Certification Body DNV-GL, awarded in 2018 for Marcon headquarters, in 2019 it was extended to the sites of Latisana and Montemarciano of San Marco Group S.p.A and to the Forlì-based Novacolor S.r.l. and, in 2020, also to the secondary production units incorporated into the San Marco Group. It obtained the Workplace safety recognition of safety management system as per the UNI INAIL (National Istitute for Insurance against Accidents at Work) model, for the Marcon headquarters.

San Marco Group received the Eurofins Indoor Air Comfort Gold certification, which is achieved if the products meet precise requirements for low VOC (Volatile Organic Compounds) emissions. In addition, the Marcon and Forlì sites have obtained REDcert^2 certification for a wide range of products, which is a certification schemes for the use of sustainable biomass in the chemical industry.

1.2 Environmental policy

San Marco Group offered water-based solutions for enamels and stains back in 1982 with the Unimarc Line and since then the pursuit has continued towards eliminating from its formulations raw materials considered hazardous to humans and the environment.

In 2010 San Marco Group has begun using the LCA methodology to understand the environmental performance of its products and to analyze their strengths and weaknesses. The holistic view of LCA convinced San Marco Group to acquire internally skills on the methodology and its application, with the ambitious goal of carrying out the LCA analysis of all the main products.

The LCA studies conducted enable San Marco Group to get a picture of its products from an environmental point of view, and to take actions of eco-design, both through actions for the improvement of its processes, and through the involvement of the supply chain in a virtuous circle. Moreover, the application of LCA in 2011 has allowed San Marco to achieve the EPD, or Environmental Product Declaration, for three products in the International EPD System.

In 2015 the thermal insulation system Marcotherm achieved the EPD.

In 2018 San Marco obtained the EPD process within the International EPD System which enables San Marco to create EPD autonomously. The EPD process is verified by a third party verifier and, internally, once a year.

1.3 Product description

The object of this EPD is a mineral base coat in powder Archi+, it has been chosen to conduct a representative product analysis, the product choice is based on production volumes. Archi+ Pietra Intonaco has been chosen as a representative product because it is the best-selling of all the products considered in the study in 2022.

The study takes into account the following products:

- Archi+ Pietra Intonaco: based on hydraulic lime with bigger grain; it is made of natural hydraulic lime, aerial lime, specific aggregates and rheological modifiers carefully selected to ensure an excellent workability.
- Archi+ Pietra: lime-based powder coating, based on hydraulic lime with stone effect; it is made of natural hydraulic lime, aerial lime, specific aggregates and rheological modifiers carefully selected to ensure an excellent workability. The particular formulation allows the attainment of top quality finishes;
- Archi+ Argilla: mineral plaster coat in powder, formulated with natural clay; it is made of clay, aerial lime, carefully selected aggregates and rheological modifiers to ensure excellent workability;
- Archi+ Tadelakt: lime-based powder coating, based on hydraulic lime; it is made of natural hydraulic lime, aerial lime, specific aggregates rheological modifiers carefully selected to ensure an excellent workability. The particular formulation allows replicating, from an aesthetic point of view, the typical finishes of Moroccan Tadelakt;
- Archi+ Fondo: Lime-based key coat in powder; it is made of hydrated lime, white cement, aggregates and rheological modifiers carefully selected to ensure an excellent workability.
- Archi+ Big: is a mineral plaster coat in powder for interiors and exteriors. It is made of well-seasoned hydrated lime putties, white cement, specific aggregates and rheological modifiers carefully selected to ensure an excellent workability. The particular grain curve of the product and the application technique ensure to achieve pleasant aesthetic effect, even with the application of one single layer.
- Archi+ Concrete: is a mineral plaster coat in powder for interiors. It is made of well-seasoned hydrated lime putties, white cement, specific aggregates and rheological modifiers carefully selected to ensure an excellent workability. The particular grain curve of Archi+ Concrete and the possibility to overlap it with one or more layers of siloxane phase, acrylsiloxane colour wash (matt), ensures to achieve cutting-edge elegant concrete effects, in line with the newest trends in the interior design.

Particularly, the most representative product is Archi+ Pietra Intonaco, the study has been referred to this product, but the results of the other products has been expressed through conversion factors.

The analysis is in line with the PCR 2019:14 v1.3.1 “Construction products” [4] and the EN 15804:2012+A2:2019/AC2021 E [5].

All Archi+ products are produced by Eurobeton, which is part of San Marco Group, in the production plant located in the municipality of Latisana, in the province of Udine, after the product has been sent to Novacolor, in the production plant in the municipality of Forlì, in the province of Forlì-Cesena, which is part of San Marco Group, for the successful distribution phase.

The UN CPC code is 374 “Plaster, lime and cement”.

The product is available white in the package of 15 kg.

1.4 Content information

The Archi+ Pietra Intonaco composition is shown in the follow tables:

		15 kg package		Declared Unit	Biogenic material		Range other products in the study	
		Weight (kg)	%	Weight (kg)	Weight (% and kg C/kg)		Weight (kg) min	Weight(kg) max
Product	Additives	0.221	1.4%	0.015	0.0013	0.13%	0.180	0.872
	Fillers	14.508	93.5%	0.967	0.0037	0.36%	13.825	14.551
	Dispersion and resins	0.271	1.7%	0.018	0.0000	0.00%	0.181	0.605
	Total bulk product	**15.000**	**96.7%**	**1.000**	**0.0051**	**0.49%**	**15.000**	
1° packaging	Paper	0.160	1.0%	0.011	0.0045	0.43%	0.160	
	Polyethylene	0.012	0.1%	0.001	0.0000	0.00%	0.012	
	Total primary packaging	**0.172**	**1.1%**	**0.012**	**0.0045**	**0.43%**	**0.172**	
3° packaging	Wood	0.341	2.2%	0.023	0.0089	0.86%	0.341	
	Polyethylene	0.006	0.0%	0.0004	0.0000	0.00%	0.006	
	Total tertiary packaging	**0.347**	**2.2%**	**0.023**	**0.0089**	**0.86%**	**0.347**	
	Total packaging	**0.519**	**3.3%**	**0.035**	**0.0134**	**1.30%**	**0.519**	
	Total	**15.519**	**100.0%**	**1.035**	**0.0185**	**1.79%**	**15.519**	

Table 1 – Archi+ Pietra Intonaco

All Archi+ products contains no SVHC (substances of very high concern) on the Candidate List published by ECHA (European Chemicals Agency) in a concentration more than 0.1 % (w/w).

2. Environmental performance declaration

2.1 Declared unit

The declared unit is 1 kg of mineral base coat in powder with packaging.

The average consumption of the product Archi+ Pietra Intonaco is 1.8-2.2 kg/m2. This value is in the technical data sheet.

The follow table show the average consumption of other Archi+ family products:

Archi+ Pietra	0.8-1.2 kg/m^2
Archi+ Argilla	1.8-2.2 kg/m^2
Archi+ Tadelakt	0.9-1.2 kg/m^2
Archi+ Fondo	0.5-0.7 kg/m^2
Archi+ Big	1.2-1.8 Kg/m^2
Archi+ Concrete	0.9-1.2 Kg/m^2

Table 2 – Archi+ average consumption

2.2 System boundaries

The present EPD is a declaration "from cradle to gate with options, modules C1-C4, module D and with optional modules" (type "b" EPD). The system boundaries include production phase (A1-A2-A3), the application phase (A5), the end of life phase (C1-C2-C3-C4) and the recovery phase (D). For completeness, the product packaging and its disposal are included.

The use (B1-B7) is excluded because they are strongly characterized by the conditions in which the system is used and also distribution phase (A4) is excluded because the data are not representative for the radical variation year by year. B1, B2, B3, B4, B5, B6 and B7 modules are not declared.

The **Production phase** for San Marco manufacturing include:

- A1 - Raw materials and their packaging: raw materials extraction, production of product components and packaging, and energy consumption (use of a mechanical mixer for liquid products, internal transfers with electric vehicles, packaging with primary packaging materials, palletisation, product storage);
- A2 – Transport: transportation of raw materials and packaging to San Marco plant;
- A3 – Manufacturing: washing operations, production and disposal of scrap, water consumption and waste treatment.

Application phase include (after the production process):

- A5 - Construction installation: product application, water dilution, waste and packaging end-of-life.

End of life phase include:

- C1 – Deconstruction demolition,
- C2 - EOL Transport: transport to landfill of the de-constructed product,
- C3 - Waste processing: collection and treatment of material flow for reuse, recycling and energy recovery. No waste processing for all Archi+ products are expected.
- C4 - Disposal: the complete andfill disposal has been assumed for the product.

The impact of the reuse and recovery phase is assessed separately:

- D - Recovery, reuse and recycling potential.

	Product stage			**Construction process stage**		**Use stage**							**End of life stage**				**Resource recovery stage**
	Raw materials	Transport	Manufacturing	Transport	Construction installation	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	De-construction demolition	Transport	Waste processing	Disposal	Reuse-Recovery-Recycling-potential
Module	A1	A2	A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
Modules declared	x	x	x	ND	x	ND	ND	ND	ND	ND	ND	ND	x	x	x	x	x
Geography	GLO	GLO	IT	-	GLO	-	-	-	-	-	-	-	GLO	GLO	GLO	GLO	GLO
Specific data used	<10%																
Variation – products	0.3-15.9%																
Variation - sites	0%																

Table 3 - Modules evaluated of the life cycle

The variation between products in GWP-GHG indicator results are greater than 10%, but the products were considered in the same study because they have very similar formulation and belong to the same product family.

2.3 Time boundaries

Primary data come from San Marco Group and refer to the year 2023. Secondary data come from the database ecoinvent v3.9.1 [6] released in 2022 and available in the LCA software used for calculations SimaPro 9.6 [7].

2.4 Geographic boundaries

The production site San Marco is located in the municipality of Latisana, in the province of Udine, after the products have been sent to site San Marco Novacolor, in the municipality of Forlì, in the province of Forlì-Cesena for the successful distribution. Since Archi+ products analyse components are mainly produced, sold and used in all the world, the study refers to distribution data. End of life scenarios of packaging refer to national data taken from the Eurostat [9] database, OECD [10] and secondary data [11].

The most significant database processes have been modified in order to make them more representative.

2.5 Life cycle boundaries

In accordance with the options of the PCR for "Construction products", the following processes are excluded from the LCA: the construction of company buildings, the production of manufacturing equipment and other capital goods, personnel activities.

The contribution of infrastructure has not been excluded from the processes that already contained it, like the ecoinvent secondary processes.

2.6 Boundaries to nature and other systems

Wastewater treatment, transport and waste treatment from the production process were included in the LCA. Wastewater from the production plant is sent to an external wastewater treatment plant.

Biogenic carbon removals and emissions are assessed with the Total Global Warming Potential (GWP-total) (including fossil GWP, biogenic GWP and land use GWP).

As required by the PCR 2019:14 v1.3.1, an additional mandatory indicator is included (GWP-GHG), adopting the carbon neutrality approach. This permits comparability with EPDs based on other PCRs, aligned with version 4.0 of the GPI [8].

2.7 Allocation and cut-off rules

As regards end-of-life allocation, the "cut-off" approach was adopted. Raw materials and manufacturing processes are included for virgin resources. No allocation is made for materials subject to recycling. The recycling process is included for the input of recycled resources. The outputs subject to recycling are considered inputs for the next life cycle.

To calculate the production consumption of 1 kg of product, the annual factory consumption and waste of San Marco in Latisana (UD) plant in 2023 were attributed on a mass basis, considering the total quantity of all products manufactured.

No cut-off rules were applied.

No co-products allocation was applied because there are not co-products.

2.8 Database and LCA software used

Ecoinvent 3.9.1., SimaPro 9.6

3. Data quality

Primary data have been used for the fundamental aspects of the study, such as energy and water consumption in production plants of San Marco or the product composition. For all processes for which primary or representative data were not available, the LCA database ecoinvent v3.9.1, cut-off by classification, was used. The processes selected from the ecoinvent database have been modified, if necessary, to make them more representative.

Data quality has been evaluated through the pedigree matrix for all processes with incidence >2% for the main LCIA categories and a qualitative analysis for remaining processes. A sensitivity analysis of proxy data has been carried out to verify the contribution to the main LCIA categories is lower than 10% as required by the reference PCR.

All inputs of the productive process have been evaluated.

For data collection and LCA calculations, the methodology described in the manual "EPD di processo San Marco" has been used.

Raw materials not available in the ecoinvent database have been modelled from the information contained in the respective safety and technical data sheets or by modifying existing ecoinvent processes of similar chemicals.

For San Marco plant electricity consumption, a residual electricity mix (Electricity, medium voltage {IT}| electricity, medium voltage, residual mix | Cut-off, S) as purchased[1].

To define the plant consumption (A3) data concerning the consumption of electricity, water and waste production in 2023 have been used.

La application stage (A5) include:

- water consumption due to the paint dilution, as reported in the technical data sheet;
- application by hand, without energy consumption;
- application scrap end of life;
- packaging material end of life.

It is assumed that 100% of the scrap is sent to landfill for disposal.

[1] The GWP-GHG value related to the electricity mix is available on request to the company

Scenario information	Unit (expressed per declared unit)	
Auxiliary materials for installation	-	
Water use	0.0004 m^3	
Energy consumption during the installation process	0.0042 kWh (Electricity. medium voltage {GLO}| market group for electricity. medium voltage | Cut-off. S)	
General waste during the installation process	Product scrap	Assumption 0,01 kg
	Primary packaging	0.01092 kg of paper and 0.00082 kg of polyethylene
	Secondary packaging	0.0227 kg of pallet and 0.0004 kg of polyethylene
Direct emissions to air	0.000067 m^3 water	

Table 4 – Installation of the products in the building

Packaging materials are disposed according to distribution data and specific disposal percentage from Eurostat [9] database, OECD [10] database and secondary data [11].

Scenario	Archi+Pietra Intonaco
UE	51.82%
Italy	22.76%
Asia-Oceania	9.36%
Africa	0.11%
USA	12.64%
South America	1.49%
Tunisia	0.00%
Morocco	0.42%
United Arab Emirates	0.00%
Saudi Arabia	1.40%
Oman	0.00%
Qatar	0.00%

Table 5 – Distribution data

Eurostat [9] data has been used for primary and tertiary packaging of Italian and European disposal scenarios. OECD data [10] has been used for Asia-Oceania, South America and USA disposal scenarios, while secondary data [11] has been used for Africa, Tunisia, Morocco, United Arab Emirates, Saudi Arabia, Oman and Qatar disposal scenario.

Scenario	Type	% Recycling	% Landfill	% Incineration
EU	Carta	83%	9%	8%
	Plastica	41%	22%	37%
	Legno	32%	37%	31%
Asia-Oceania	-	29%	23%	48%
South America-USA	-	23%	67%	10%
Italy	Carta	85%	9%	6%
	Plastica	48%	11%	41%
	Legno	23%	67%	10%
Africa	-	8%	92%	-
Saudi Arabia	-	15%	85%	-
United Arab Emirates	-	22%	78%	-
Morocco	-	8%	92%	-
Oman	-	-	100%	-
Qatar	-	3%	97%	-
Tunisia	-	4%	96%	-

Table 6 – Disposal data

It is assumed that Archi+ products are sent to landfill for disposal (C3-C4). Transport to disposal is assumed to be 50 km (C2) and 100 km for packaging and product disposal.

The C3 module is empty because no treatment of materials for reuse, recycling and energy recovery are expected. The C1 module is zero because the environmental impacts for demolition associated to the paint are low.

Processes	Unità (espressa in funzione dell'unità dichiarata)
Collection process	0.99 kg of product after application (0.01 kg is application scrap in A5)
Disposal type	0.99 kg to landfill
Assumption for scenario development	Distribution percentage have been used and the landfill scenario has been subdivided in two scenarios. european and global
Assumption transportation	100 km (Transport. freight. lorry 16-32 metric ton. EURO4 {RER}| market for transport. freight. lorry 16-32 metric ton. EURO4 | Cut-off. S e Transport. freight. lorry 16-32 metric ton. EURO4 {RoW}| market for transport. freight. lorry 16-32 metric ton. EURO4 | Cut-off. S)

Table 7 – End of life

As regards module D, benefits and impacts related to the recovery of the packaging paper, wood and polypropylene have been considered respectively in terms of use of recycled materials in place of virgin materials and energy recovery due to incineration with energy recovery. It is assumed that a recycled material replaces a material with the same origin. For wood and paper a factor equal to 1 is assumed. There are no secondary fuels. Product and packaging are assumed to not produce landfill gas.

4. Environmental impact

4.1 Environmental impact indicators

The following environmental impact indicators are considered, according to PCR 2019:14 v1.3.1 and UNI EN EN 15804:2012+A2:2019/AC2021 E.

Nr.	Core environmental impact indicators	Unit
1.1	Global Warming Potential - total (GWP-total)	kg CO2 eq.
1.2	Global Warming Potential - fossil fuels (GWP-fossil)	kg CO2 eq.
1.3	Global Warming Potential - biogenic (GWP-biogenic)	kg CO2 eq.
1.4	Global Warming Potential - land use and land use change (GWP-luluc)	kg CO2 eq.
1.5	Depletion potential of the stratospheric ozone layer (ODP)	kg CFC-11 eq.
1.6	Acidification potential, Accumulated Exceedance (AP)	mol H+ eq.
1.8	Eutrophication potential - freshwater (EP-freshwater)	kg P eq
1.9	Eutrophication potential - marine (EP-marine)	kg N eq.
1.10	Eutrophication potential - terrestrial (EP-terrestrial)	mol N eq.
1.11	Photochemical Ozone Creation Potential (POCP)	kg NMVOC eq.
1.12	Abiotic depletion potential - non-fossil resources (ADPE)	kg Sb eq.
1.13	Abiotic depletion potential - fossil resources (ADPF)	MJ
1.14	Water (user) deprivation potential (WDP)	m3 world eq. deprived
	Additional mandatory environmental impact indicators	
2.1.	Global Warming Potential (GWP-GHG)	kg CO2 eq.
	Additional environmental impact indicators	
3.1	Particulate Matter emissions (PM)	Disease incidence
3.2	Ionizing radiation, human health (IRP)	kBq U235 eq.
3.3	Eco-toxicity - freshwater (ETP-fw)	CTUe
3.4	Human toxicity, cancer effect (HTP-c)	CTUh
3.5	Human toxicity, non-cancer effects (HTP-nc)	CTUh
3.6	Land use related impacts/Soil quality (SQP)	dimensionless
	Indicators describing resource use	
4.1	Use of renewable primary energy as energy carrier (PERE)	MJ
4.2	Use of renewable primary energy resources used as raw materials (PERM)	MJ
4.3	Total use of renewable primary energy (PERT)	MJ
4.4	Use of non renewable primary energy as energy carrier (PENRE)	MJ

4.5	Use of non renewable primary energy resources used as raw materials (PENRM)	MJ
4.6	Total use of non renewable primary energy resource (PENRT)	MJ
4.7	Use of secondary material (SM)	kg
4.8	Use of renewable secondary fuels (RSF)	MJ
4.9	Use of non renewable secondary fuels (NRSF)	MJ
4.10	Net use of fresh water (FW)	m3
	Environmental information describing waste categories	
5.1	Hazardous waste disposed (HWD)	kg
5.2	Non hazardous waste disposed (NHWD)	kg
5.3	Radioactive waste disposed (RWD)	kg
	Environmental information describing output flows	
6.1	Components for re-use (CRU)	kg
6.2	Materials for recycling (MFR)	kg
6.3	Materials for energy recovery (MER)	kg
6.4	Exported energy (EE), electricity	MJ
6.5	Exported energy (EE), thermal	MJ
	Biogenic carbon content	
7.1	Biogenic carbon content in product	kg C
7.2	Biogenic carbon content in accompanying packaging	kg C

Table 8 - Environmental indicators included in the analysis.

4.2 Environmental impact assessment of Archi+ Pietra Intonaco

Potential environmental impact – mandatory indicators according to EN 15804

Result for declared unit								
Indicator	**Unit**	**A1-A2-A3**	**A5**	**C1**	**C2**	**C3**	**C4**	**D**
GWP-total	kg CO2 eq.	4.53E-01	5.40E-02	0.00E+00	9.31E-03	0.00E+00	2.46E-02	7.61E-03
GWP-fossil	kg CO2 eq.	4.96E-01	5.71E-03	0.00E+00	9.30E-03	0.00E+00	2.46E-02	-4.07E-03
GWP-biogenic	kg CO2 eq.	-4.33E-02	4.83E-02	0.00E+00	7.12E-06	0.00E+00	1.77E-05	1.17E-02
GWP-luluc	kg CO2 eq.	3.31E-04	7.23E-06	0.00E+00	4.63E-06	0.00E+00	1.29E-05	-1.41E-05
ODP	kg CFC 11 eq.	8.21E-09	1.03E-10	0.00E+00	1.89E-10	0.00E+00	5.53E-10	-7.10E-11
AP	mol H+eq.	1.44E-03	2.30E-05	0.00E+00	3.91E-05	0.00E+00	1.24E-04	-1.23E-05
EP-freshwater	kg P eq	6.78E-05	1.61E-06	0.00E+00	6.81E-07	0.00E+00	1.86E-06	-9.26E-06
EP-marine	kg N eq.	4.02E-04	9.56E-06	0.00E+00	1.48E-05	0.00E+00	4.69E-05	-6.12E-06
EP-terrestrial	mol N eq.	4.26E-03	6.24E-05	0.00E+00	1.58E-04	0.00E+00	5.02E-04	-3.22E-05
POCP	kg NMVOC eq.	1.68E-03	1.97E-05	0.00E+00	5.60E-05	0.00E+00	1.77E-04	-1.97E-05
ADP-minerals & metals*	kg Sb eq.	1.62E-06	7.60E-09	0.00E+00	3.00E-08	0.00E+00	6.84E-08	1.02E-09
ADP-fossil*	MJ	5.93E+00	5.97E-02	0.00E+00	1.32E-01	0.00E+00	4.15E-01	-6.44E-02
WDP*	m3eq	8.32E-02	1.71E-02	0.00E+00	5.50E-04	0.00E+00	7.72E-03	-1.52E-04
Acronyms		GWP-fossil = Global Warming Potential fossil fuels; GWP-biogenic = Global Warming Potential biogenic; GWP-luluc = Global Warming Potential land use and land use change; ODP = Depletion potential of the stratospheric ozone layer; AP = Acidification potential, Accumulated Exceedance; EP-freshwater = Eutrophication potential, fraction of nutrients reaching freshwater end compartment; EP-marine = Eutrophication potential, fraction of nutrients reaching marine end compartment; EP-terrestrial = Eutrophication potential, Accumulated Exceedance; POCP = Formation potential of tropospheric ozone; ADP-minerals&metals = Abiotic depletion potential for non-fossil resources; ADP-fossil = Abiotic depletion for fossil resources potential; WDP = Water (user) deprivation potential, deprivation-weighted water consumption						

Table 9 - Results for declared unit

** Disclaimer: The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experience with the indicator.*

The estimated impact results are only relative statements, which do not indicate the endpoints of the impact categories, exceeding threshold values, safety margins and/or risks.

Potential environmental impact – additional indicators

Results for declared unit								
Indicator	**Unit**	**A1-A2-A3**	**A5**	**C1**	**C2**	**C3**	**C4**	**D**
GWP-GHG[2]	kg CO2 eq.	4.97E-01	6.40E-03	0.00E+00	9.31E-03	0.00E+00	2.46E-02	-4.19E-03
PM	Disease incidence	1.71E-08	2.62E-10	0.00E+00	7.62E-10	0.00E+00	2.52E-09	-5.72E-11
IRP**	kBq U235 eq.	1.83E-02	4.88E-04	0.00E+00	1.61E-04	0.00E+00	4.16E-04	-5.36E-04
ETP-fw*	CTUe	1.86E+00	2.03E-02	0.00E+00	6.72E-02	0.00E+00	2.05E-01	-2.47E-03
HTP-c*	CTUh	1.60E-10	3.64E-12	0.00E+00	4.25E-12	0.00E+00	1.11E-11	-1.92E-12
HTP-nc*	CTUh	3.00E-09	5.70E-11	0.00E+00	9.37E-11	0.00E+00	2.20E-10	1.01E-11
SQP*	dimensionless	7.21E+00	2.34E-02	0.00E+00	7.89E-02	0.00E+00	4.55E-01	-9.57E-01
Acronyms		GWP-GHG = Global Warming Potential neutral; PM = Potential incidence of disease due to PM emissions; IRP = Potential human exposure efficiency relative to U235, ETP,fw = Potential comparative toxic unit for ecosystems; HTP-c = Potential comparative toxic unit for effects; HTP-nc = Potential comparative toxic unit for humans; SQP = Potential soil quality index.						

Table 10 - Result for declared unit - additional indicator

** Disclaimer: The results of this environmental impact indicator shall be used with care as the uncertainties of these results are high or as there is limited experience with the indicator.*

*** Disclaimer: This impact category deals mainly with the eventual impact of low dose ionizing radiation on human health of the nuclear fuel cycle. It does not consider effects due to possible nuclear accidents, occupational exposure nor due to radioactive waste disposal in underground facilities, Potential ionizing radiation from the soil, from radon and from some construction materials is also not measured by this indicator.*

[2]The indicator includes all greenhouse gases included in GWP-total but excludes biogenic carbon dioxide uptake and emissions and biogenic carbon stored in the product

Use of resources

Results for declared unit								
Indicator	**Unit**	**A1-A2-A3**	**A5**	**C1**	**C2**	**C3**	**C4**	**D**
PERE	MJ	8.87E-01	5.33E-02	0.00E+00	1.96E-03	0.00E+00	0.00E+00	0.00E+00
PERM	MJ	4.60E-01	-3.01E-01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PERT	MJ	1.35E+00	-2.48E-01	0.00E+00	1.96E-03	0.00E+00	0.00E+00	0.00E+00
PENRE	MJ	5.87E+00	1.09E-02	0.00E+00	1.32E-01	0.00E+00	0.00E+00	0.00E+00
PENRM	MJ.	5.20E-02	-3.77E-02	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PENRT	MJ	5.93E+00	-2.68E-02	0.00E+00	1.32E-01	0.00E+00	0.00E+00	0.00E+00
SM	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
RSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
NRSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
FW	m3	2.57E-03	4.32E-04	0.00E+00	1.89E-05	0.00E+00	1.97E-04	-5.44E-05
Acronyms		PERE = Use of renewable primary energy excluding renewable primary energy resources used as raw materials; PERM = Use of renewable primary energy resources used as raw materials; PERT = Total use of renewable primary energy resources; PENRE = Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials; PENRM = Use of non-renewable primary energy resources used as raw materials; PENRT = Total use of non-renewable primary energy re-sources; SM = Use of secondary material; RSF = Use of renewable secondary fuels; NRSF = Use of non-renewable secondary fuels; FW = Use of net fresh water						

Table 11 - Use of resources

Waste production and output flows

Waste production

Results for declared unit								
Indicator	**Unit**	**A1-A2-A3**	**A5**	**C1**	**C2**	**C3**	**C4**	**D**
Hazardous waste disposed	kg	2.41E-05	1.73E-07	0.00E+00	8.47E-07	0.00E+00	2.49E-06	-8.82E-07
Non-hazardous waste disposed	kg	1.12E-01	2.31E-02	0.00E+00	6.46E-03	0.00E+00	1.00E+00	-7.97E-04
Radioactive waste disposed	kg	4.61E-06	1.19E-07	0.00E+00	3.89E-08	0.00E+00	9.99E-08	-1.36E-07

Table 12 - Waste production

Output flows

Result for declared unit								
Indicator	**Unit**	**A1-A2-A3**	**A5**	**C1**	**C2**	**C3**	**C4**	**D**
Components for re-use	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Material for recycling	kg	0.00E+00	8.23E-03	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Materials for Energy recovery	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Exported energy, electricity	MJ	0.00E+00	1.48E-02	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Exported energy, thermal	MJ	0.00E+00	2.84E-02	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

Table 13 - Output flows

Information on biogenic carbon content

Results for declared unit		
BIOGENIC CARBON CONTENT	**Unit**	**QUANTITY**
Biogenic carbon content in product	kg C	Not significant (<5%)
Biogenic carbon content in packaging	kg C	Not significant (<5%)

Table 14 - Biogenic carbon content

Note: 1 kg biogenic carbon is equivalent to 44/12 kg CO2.

5. Additional environmental information

5.1 Conversion factors

In accordance with PCR 2019:14 v1.3.1 "Construction products", conversion factors may be included to converting the declared results of a product group for specific products within the group.

To evaluate the environmental performance of all the variants, conversion factors were extrapolated to be used for the impacts calculation:

- Archi + Pietra;
- Archi+ Argilla;
- Archi+ Tadelakt;
- Archi+ Fondo;
- Arhi+ Big;
- Archi+ Concrete.

To extrapolate the conversion factors, an LCA model has been created for each variant. The impact of each product was related to the reference product for each module.

To determinate the environmental impacts associated with each product, the conversion factors must be multiplied by the impacts of the reference product. Each environmental indicator value for each phase shall be calculated using the following formulas:

$$y = a * x$$

Where:

- *y is the impact chosen category;*
- *x is the impact chosen category of reference product;*
- *a is the conversion factor;*

The following tables shown the conversion factors for each product and for each module considered.

Factors for A1-A3 modules	Archi+ Argilla	Archi+ Big	Archi+ Concrete	Archi+ Fondo	Archi+ Pietra	Archi+ Tadelakt
GWP-total	8.39E-01	9.96E-01	9.79E-01	1.17E+00	1.00E+00	1.10E+00
GWP-fossil	8.59E-01	9.97E-01	9.85E-01	1.16E+00	1.00E+00	1.10E+00
GWP-biogenic	1.07E+00	1.01E+00	1.05E+00	9.96E-01	1.04E+00	1.04E+00
GWP-luluc	9.21E-01	9.77E-01	1.29E+00	1.09E+00	1.26E+00	1.33E+00
ODP	7.83E-01	1.01E+00	1.06E+00	1.14E+00	1.01E+00	1.11E+00
AP	7.33E-01	9.79E-01	1.02E+00	1.04E+00	1.02E+00	1.01E+00
EP-freshwater	7.81E-01	9.86E-01	1.06E+00	1.14E+00	1.04E+00	1.10E+00
EP-marine	7.67E-01	9.49E-01	9.63E-01	1.02E+00	1.03E+00	1.02E+00
EP-terrestrial	7.51E-01	9.43E-01	9.54E-01	1.01E+00	1.03E+00	1.01E+00
POCP	8.32E-01	9.84E-01	9.88E-01	1.06E+00	1.02E+00	1.04E+00
ADP-minerals&metals	5.94E-01	1.12E+00	1.19E+00	1.14E+00	9.83E-01	1.07E+00
ADP-fossil	8.37E-01	1.26E+00	1.35E+00	1.25E+00	1.01E+00	1.12E+00
WDP	1.01E+00	1.07E+00	1.28E+00	1.43E+00	1.12E+00	1.16E+00
GWP-GHG	8.59E-01	9.97E-01	9.85E-01	1.16E+00	1.00E+00	1.10E+00
PM	8.81E-01	1.08E+00	1.09E+00	1.11E+00	1.03E+00	1.08E+00
IRP	7.49E-01	9.68E-01	1.06E+00	1.25E+00	1.03E+00	1.13E+00
ETP-fw	9.47E-01	1.01E+00	1.01E+00	1.10E+00	1.05E+00	1.12E+00
HTP-c	8.73E-01	1.01E+00	1.03E+00	1.07E+00	1.03E+00	1.07E+00
HTP-nc	7.66E-01	1.01E+00	1.05E+00	1.15E+00	1.06E+00	1.13E+00
SQP	9.44E-01	9.99E-01	1.01E+00	1.03E+00	1.01E+00	1.02E+00
PERE	8.52E-01	9.94E-01	1.05E+00	1.08E+00	1.00E+00	1.06E+00
PERM	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
PERT	9.03E-01	9.96E-01	1.04E+00	1.05E+00	1.00E+00	1.04E+00
PENRE	8.36E-01	1.27E+00	1.35E+00	1.25E+00	1.02E+00	1.12E+00
PENRM	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
PENRT	8.37E-01	1.26E+00	1.35E+00	1.25E+00	1.02E+00	1.12E+00
SM	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
RSF	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
NRSF	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
FW	9.60E-01	1.06E+00	1.24E+00	1.37E+00	1.10E+00	1.15E+00
Hazardous waste disposed	9.56E-01	1.03E+00	1.01E+00	1.06E+00	1.03E+00	1.09E+00
Non-hazardous waste disposed	9.71E-01	1.02E+00	9.79E-01	1.09E+00	1.03E+00	1.05E+00
Radioactive waste disposed	7.42E-01	9.66E-01	1.06E+00	1.25E+00	1.03E+00	1.13E+00
Component for re-use	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
Materials for recycling	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
Materials for energy recovery	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
Exported energy, electricity	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
Exported energy, thermal	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00

Table 15 – Conversion factors for calculating the potential environmental impacts related to A1-A3 of products considered in the study

Factors for A5 module	Archi+ Argilla	Archi+ Big	Archi+ Concrete	Archi+ Fondo	Archi+ Pietra	Archi+ Tadelakt
GWP-total	1.00E+00	1.01E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
GWP-fossil	1.01E+00	1.05E+00	9.96E-01	1.02E+00	1.01E+00	9.93E-01
GWP-biogenic	1.00E+00	1.01E+00	1.00E+00	1.00E+00	1.00E+00	1.01E+00
GWP-luluc	9.83E-01	9.94E-01	9.96E-01	9.96E-01	9.95E-01	9.98E-01
ODP	8.94E-01	9.07E-01	8.98E-01	9.64E-01	9.85E-01	8.90E-01
AP	9.82E-01	1.02E+00	9.92E-01	1.00E+00	1.00E+00	9.97E-01
EP-freshwater	9.80E-01	1.00E+00	9.89E-01	9.97E-01	9.96E-01	9.96E-01
EP-marine	9.79E-01	9.65E-01	1.10E+00	9.87E-01	9.77E-01	1.05E+00
EP-terrestrial	9.86E-01	1.03E+00	9.84E-01	1.00E+00	1.01E+00	9.87E-01
POCP	9.85E-01	1.01E+00	9.90E-01	9.99E-01	1.01E+00	9.83E-01
ADP-minerals&metals	9.56E-01	9.87E-01	9.77E-01	9.90E-01	9.90E-01	9.85E-01
ADP-fossil	9.83E-01	9.97E-01	1.01E+00	9.97E-01	9.93E-01	1.01E+00
WDP	8.08E-01	8.48E-01	8.51E-01	9.39E-01	9.61E-01	8.51E-01
GWP-GHG	1.00E+00	1.03E+00	1.03E+00	1.01E+00	1.00E+00	1.01E+00
PM	9.80E-01	1.01E+00	1.01E+00	9.98E-01	9.95E-01	1.01E+00
IRP	9.82E-01	9.80E-01	9.82E-01	9.93E-01	9.97E-01	9.79E-01
ETP-fw	9.90E-01	1.15E+00	1.02E+00	1.03E+00	9.96E-01	1.11E+00
HTP-c	9.43E-01	8.74E-01	8.37E-01	9.62E-01	1.04E+00	7.44E-01
HTP-nc	9.65E-01	1.22E+00	9.89E-01	1.04E+00	9.97E-01	1.12E+00
SQP	9.80E-01	9.68E-01	1.10E+00	9.89E-01	9.72E-01	1.06E+00
PERE	1.02E+00	1.59E+00	8.37E-01	1.12E+00	1.10E+00	1.09E+00
PERM	1.03E+00	1.07E+00	7.62E-01	1.02E+00	1.06E+00	8.50E-01
PERT	1.03E+00	9.53E-01	7.46E-01	9.98E-01	1.05E+00	7.99E-01
PENRE	1.08E+00	1.13E+00	9.41E-01	1.08E+00	1.09E+00	8.75E-01
PENRM	1.05E+00	1.02E+00	8.53E-01	1.04E+00	1.06E+00	8.44E-01
PENRT	1.04E+00	9.72E-01	8.17E-01	1.02E+00	1.05E+00	8.32E-01
SM	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
RSF	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
NRSF	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
FW	8.13E-01	8.51E-01	8.52E-01	9.40E-01	9.63E-01	8.51E-01
Hazardous waste disposed	9.82E-01	1.01E+00	1.03E+00	1.00E+00	9.89E-01	1.03E+00
Non-hazardous waste disposed	9.74E-01	9.44E-01	1.23E+00	9.81E-01	9.42E-01	1.15E+00
Radioactive waste disposed	9.82E-01	9.80E-01	9.82E-01	9.93E-01	9.97E-01	9.78E-01
Component for re-use	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
Materials for recycling	1.03E+00	9.05E-01	7.57E-01	9.93E-01	1.02E+00	8.50E-01
Materials for energy recovery	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
Exported energy, electricity	1.04E+00	1.74E+00	9.05E-01	1.17E+00	1.09E+00	1.23E+00
Exported energy, thermal	1.04E+00	1.67E+00	8.89E-01	1.15E+00	1.10E+00	1.18E+00

Table 16 – Conversion factors for calculating the potential environmental impacts related to A5 of products considered in the study

The potential environmental impacts related to C1, C2 and C3 are the same of representative product Archi+ Pietra Intonaco.

Factors for C4 module	Archi+ Argilla	Archi+ Big	Archi+ Concrete	Archi+ Fondo	Archi+ Pietra	Archi+ Tadelakt
GWP-total	1.00E+00	1.01E+00	1.00E+00	1.00E+00	9.99E-01	1.01E+00
GWP-fossil	1.00E+00	1.01E+00	1.00E+00	1.00E+00	9.99E-01	1.01E+00
GWP-biogenic	1.01E+00	7.89E-01	8.34E-01	9.58E-01	1.04E+00	7.22E-01
GWP-luluc	9.98E-01	1.02E+00	1.02E+00	1.00E+00	9.95E-01	1.03E+00
ODP	1.00E+00	9.30E-01	9.45E-01	9.86E-01	1.01E+00	9.08E-01
AP	9.99E-01	1.02E+00	1.01E+00	1.00E+00	9.96E-01	1.02E+00
EP-freshwater	9.97E-01	1.05E+00	1.04E+00	1.01E+00	9.91E-01	1.06E+00
EP-marine	9.99E-01	1.01E+00	1.01E+00	1.00E+00	9.98E-01	1.01E+00
EP-terrestrial	9.99E-01	1.01E+00	1.01E+00	1.00E+00	9.98E-01	1.01E+00
POCP	1.00E+00	9.99E-01	9.99E-01	1.00E+00	1.00E+00	9.99E-01
ADP-minerals&metals	1.00E+00	1.00E+00	1.00E+00	1.00E+00	9.99E-01	1.01E+00
ADP-fossil	1.00E+00	1.00E+00	1.00E+00	1.00E+00	9.99E-01	1.00E+00
WDP	1.00E+00	1.01E+00	1.00E+00	1.00E+00	9.99E-01	1.01E+00
GWP-GHG	1.00E+00	1.01E+00	1.00E+00	1.00E+00	9.99E-01	1.01E+00
PM	1.00E+00	1.00E+00	1.00E+00	1.00E+00	9.99E-01	1.01E+00
IRP	1.01E+00	8.90E-01	9.13E-01	9.78E-01	1.02E+00	8.55E-01
ETP-fw	9.98E-01	1.03E+00	1.02E+00	1.01E+00	9.94E-01	1.04E+00
HTP-c	1.00E+00	1.00E+00	1.00E+00	1.00E+00	9.99E-01	1.01E+00
HTP-nc	9.99E-01	1.01E+00	1.01E+00	1.00E+00	9.98E-01	1.01E+00
SQP	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
PERE	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
PERM	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
PERT	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
PENRE	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
PENRM	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
PENRT	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
SM	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
RSF	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
NRSF	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
FW	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
Hazardous waste disposed	1.00E+00	1.01E+00	1.01E+00	1.00E+00	9.99E-01	1.01E+00
Non-hazardous waste disposed	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
Radioactive waste disposed	1.01E+00	8.84E-01	9.09E-01	9.77E-01	1.02E+00	8.47E-01
Component for re-use	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
Materials for recycling	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
Materials for energy recovery	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
Exported energy, electricity	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
Exported energy, thermal	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00

Table 17 – Conversion factors for calculating the potential environmental impacts related to C4 of products considered in the study

Factors for D module	Archi+ Argilla	Archi+ Big	Archi+ Concrete	Archi+ Fondo	Archi+ Pietra	Archi+ Tadelakt
GWP-total	1.00E+00	1.45E-02	5.80E-01	7.98E-01	8.16E-01	2.10E-01
GWP-fossil	1.07E+00	2.21E+00	9.68E-01	1.27E+00	1.48E+00	1.57E+00
GWP-biogenic	1.03E+00	7.82E-01	7.15E-01	9.63E-01	1.05E+00	6.85E-01
GWP-luluc	1.05E+00	1.26E+00	7.48E-01	1.06E+00	1.30E+00	9.26E-01
ODP	1.01E+00	1.54E+00	8.58E-01	1.12E+00	8.80E-01	9.98E-01
AP	1.08E+00	2.46E+00	1.00E+00	1.33E+00	1.69E+00	1.76E+00
EP-freshwater	1.03E+00	9.14E-01	7.29E-01	9.89E-01	1.11E+00	7.16E-01
EP-marine	1.05E+00	1.45E+00	8.22E-01	1.11E+00	1.33E+00	1.08E+00
EP-terrestrial	1.07E+00	2.13E+00	9.42E-01	1.25E+00	1.59E+00	1.54E+00
POCP	1.05E+00	1.50E+00	8.24E-01	1.12E+00	1.33E+00	1.10E+00
ADP-minerals&metals	1.03E+00	-1.53E+00	5.05E-02	4.47E-01	9.21E-01	-9.55E-01
ADP-fossil	1.05E+00	2.00E+00	9.48E-01	1.22E+00	1.31E+00	1.42E+00
WDP	3.88E-01	1.32E+01	7.45E+00	4.05E+00	1.28E-01	1.21E+01
GWP-GHG	1.06E+00	2.17E+00	9.59E-01	1.26E+00	1.47E+00	1.54E+00
PM	1.17E+00	3.90E+00	1.19E+00	1.64E+00	2.56E+00	2.74E+00
IRP	9.96E-01	1.84E+00	9.86E-01	1.18E+00	8.52E-01	1.23E+00
ETP-fw	1.18E+00	4.71E+00	1.28E+00	1.80E+00	2.54E+00	3.11E+00
HTP-c	1.04E+00	1.26E+00	7.74E-01	1.06E+00	1.23E+00	9.25E-01
HTP-nc	9.10E-01	-2.67E+00	8.15E-02	2.11E-01	-1.35E-01	-1.68E+00
SQP	1.01E+00	8.17E-01	7.62E-01	9.72E-01	1.04E+00	7.02E-01
PERE	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
PERM	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
PERT	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
PENRE	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
PENRM	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
PENRT	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
SM	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
RSF	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
NRSF	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
FW	1.00E+00	1.48E+00	9.82E-01	1.12E+00	1.06E+00	1.17E+00
Hazardous waste disposed	1.03E+00	9.56E-01	7.39E-01	9.98E-01	1.11E+00	7.42E-01
Non-hazardous waste disposed	1.04E+00	9.15E-01	6.54E-01	9.84E-01	1.22E+00	6.56E-01
Radioactive waste disposed	9.93E-01	1.79E+00	9.79E-01	1.17E+00	8.24E-01	1.20E+00
Component for re-use	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
Materials for recycling	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
Materials for energy recovery	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
Exported energy, electricity	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00
Exported energy, thermal	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00	1.00E+00

Table 18 – Conversion factors for calculating the potential environmental impacts related to D of products considered in the study

Results for declared unit							
BIOGENIC CARBON CONTENT	**Unit**	**Archi+ Argilla**	**Archi+ Big**	**Archi+ Concrete**	**Archi+ Fondo**	**Archi+ Pietra**	**Archi+ Tadelakt**
Biogenic carbon content in product	kg C	Not significant (<5%)	Not significant (<5%)	Not significant (<5%)	Not significant (<5%)	Not significant (<5%)	Not significant (<5%)
Biogenic carbon content in packaging	kg C	Not significant (<5%)	Not significant (<5%)	Not significant (<5%)	Not significant (<5%)	Not significant (<5%)	Not significant (<5%)

Table 19 - Biogenic carbon content

5.2 Certification

All Archi+ products are certified “Indoor Air Comfort Gold”. This certification ensures that low product emission requirements are fulfilled and it is a sign of the applicant’s focus on quality and contribution to a healthy indoor environment.

Licence number: IACG-442-01-05-2022, Licence expiry data: 2027-11-03.

Differences versus previous version

2023-11-06 Version 1

2024-11-22 Version 2

New verification: Two products have been integrated into the study (Archi+ Big and Archi+ Concrete). The formulation of products changed, particularly AO009 raw material, resulting in change to the impacts. In addition, editorial changes have been made.

6. General information

6.1 San Marco Group Spa information

The Life Cycle Assessment (LCA) study and this EPD were created by the Product Safety Department of San Marco Group SpA, in collaboration with 2B Srl (www.to-be.it).

The company contact details are:

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6.2 Programme information

Programme EPD:	The International EPD® System
Address	EPD International AB Box 210 60 SE-100 31 Stockholm Sweden
Website	www.environdec.com
E-mail	info@environdec.com

Accountabilities for PCR, LCA and independent, third-party verification
Product Category Rules (PCR)
ISO standard ISO 21930 and CEN standard EN 15804 serve as the core Product Category Rules (PCR)
Product category rules (PCR): PCR 2019:14 Construction products, version 1.3.1
PCR review was conducted by: The Technical Committee of the International EPD® System. See www.environdec.com/TC for a list of members. Review chair: Claudia A. Peña, University of Concepción, Chile. The review panel may be contacted via the Secretariat www.environdec.com/contact.
Life cycle assessment (LCA)
LCA accountability: 2B Srl
Third-party verification
Independent third-party verification of the declaration and data, according to ISO 14025:2006 via: ☑ EPD verification by EPD Process Certification* Internal auditor: Leo Breedveld, 2B Srl Third-party verification: CSQA Certificazioni Srl is an approved certification body accountable for third-party verification Third-party verifier is accredited by: Accredia (004H) *For EPD Process Certification, an accredited certification body certifies and reviews the management process and verifies EPDs published on a regular basis. For details about third-party verification procedure of the EPDs, see the GPI.
Procedure for follow-up of data during EPD validity involves third-party verifier? ☑ Yes ☐ No

San Marco Group SpA is the exclusive owner of the EPD and is responsible for its content.

EPDs within the same product category but registered in different EPD programmes may not be comparable. For two EPDs to be comparable, they must be based on the same PCR (including the same version number) or be based on fully-aligned PCRs or versions of PCRs; cover products with identical functions, technical performances and use (e.g. identical declared/functional units); have equivalent system boundaries and descriptions of data; apply equivalent data quality requirements, methods of data collection, and allocation methods; apply identical cut-off rules and impact assessment methods (including the same version of characterisation factors); have equivalent content declarations; and be valid at the time of comparison.

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